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Hybrid Explicit-Implicit FEM for Porous Media Multiphase Flow with Possible Solid-Phase Decomposition

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Abstract

Multiphase flow in porous media is ubiquitous in physical processes, yet modeling it consistently remains difficult, and sometimes it can be coupled with solid-phase decomposition and phase change, such as in hydrate dissociation or internal erosion processes. Recent code comparison studies have highlighted this difficulty, revealing clear inconsistencies in numerical results across different research groups for the same benchmark problem. This paper presents a new, reliable benchmark test and a hybrid explicit-implicit finite element method adaptable to various scenarios. In our mathematical framework, the solid decomposition is described by a rate equation for porosity that depends on the fluid pressure, and the phase change is modeled via mass source terms. The hybrid explicit-implicit finite element method features a novel three-stage updating strategy, which incorporates an artificial diffusion term and carefully selects the transport equation for the final saturation update. Validation results demonstrate that our proposed method achieves substantial agreement with those of the fully implicit finite volume method, confirming its reliability. Furthermore, our analysis confirms that the saturation update must use the transport equation of the incompressible fluid phase, and that the artificial diffusion term is critical for capturing physically correct saturation profiles, even when advection is not dominant. Overall, this work provides a consistent and effective tool for simulating complex multiphase flow scenarios and serves as a valuable complement to future benchmark studies.

Keywords: finite element; finite volume; multiphase flow; hybrid method; solid decomposition and phase change

1. Introduction

Multiphase flow in porous media occurs across a wide range of systems, both natural and engineered, including oil and gas reservoirs [1–3], methane hydrate-bearing sediments (MHBS) [4–9], and rocks undergoing solid dissolution or internal erosion [10–12]. Although the governing equations are largely formulated based on mass conservation principles [13], which are therefore similar among different systems, the numerical methods used to solve these systems vary substantially, as discussed below.

In reservoir engineering, the finite volume method (FVM) (sometimes referred to as the integral finite difference method) has become the dominant numerical approach in modern reservoir simulators [14–17], although alternative techniques such as the mimetic finite difference (MFD) [18] and discontinuous Galerkin (DG) methods [19,20] are also in use. The standard finite element method (FEM) has been proven to be inadequate for reservoir

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simulation, particularly when capturing the sharp saturation shocks characteristic of two-phase displacement processes or thermal convection processes [21]. In the field of methane hydrates, a wide variety of numerical simulation tools exist (e.g., see White et al. [22] and Zhang et al. [23]), yet no single dominant numerical scheme has emerged to date. Indeed, as illustrated by Figures 10–12 in Song et al. [5] and the benchmark comparisons in White et al. [22], many of these methods often fail to consistently produce accurate results even for simpler benchmark problems where a unique true solution should exist (although such solutions may not have analytical forms). This inconsistency, wherein a code that performs adequately for complex problems might yield erroneous results for simpler cases, ultimately hampers further progress in numerical simulation techniques of MHBS. For solid dissolution or internal erosion processes, the standard FEM is more commonly employed than alternative methods [10,11].

In summary, these scenarios share several common features. The pore spaces are filled by multiple fluid phases such as water, gas, oil, and even flowing or eroded solids [12], and their governing equations are built on mass conservation principles, typically comprising an accumulation term, a Darcy flux term, and a source/sink term [13]. In solid dissolution problems, the gradual removal of the solid phase through dissolution or erosion leads to an increase in porosity [10]. Similarly, in methane hydrate-bearing sediments, the solid matrix is treated as a composite of pure solid grains and hydrate; as the hydrate gradually dissociates, the composite solid fraction ϕ^{cs} decreases and the porosity (defined as $\phi^{f} = 1 - \phi^{cs}$) increases, mimicking the characteristics of solid dissolution or internal erosion. Generally speaking, all of the solid-phase changes described above can be referred to as "solid-phase decomposition", a term that will be used consistently in the subsequent discussions. Another issue is that many FEM-based and MPM-based methods developed for modeling methane hydrates [23–26] have not been applied to two-phase immiscible displacement problems [27,28], even though the underlying governing equations are essentially the same. This discrepancy raises questions about the robustness and general applicability of the developed numerical schemes.

Therefore, in this study, we propose a hybrid explicit-implicit FEM suitable for both two-phase immiscible fluid displacement and solid decomposition problems. The paper is organized as follows: In Section 2, we outline the model assumptions and governing equations for multiphase flow and porosity changes caused by solid decomposition. Next, in Section 3, the hybrid explicit-implicit FEM is presented. We validate the numerical solution in Section 4 by comparing it with results obtained from the finite volume method (FVM) or analytical solutions derived via the method of characteristics. Finally, conclusions are drawn in Section 5.

2. Mathematical Model

2.1. Model Assumptions

Before the formal description of the mathematical model, the following assumptions are given beforehand:

1. The wetting phase fluid intrinsic density ρ_w is considered constant, which takes the common value of 1000 kg/m^3.
2. The gas obeys the ideal gas law. Here, instead of the subscript "g", we use "nw" to denote the gas phase, which makes the description also suitable for the Buckley-Leverett problem. Therefore, the gas intrinsic density $\rho_{nw} = pM_g/(R\,\Theta)$, where p is the gas pressure; M_g is the gas molar mass, which takes $M_g = 16.042$ g/mol for methane, $R = 8.31446$ J/mol/K is the universal gas constant; and Θ is the temperature, which takes $\Theta = 275.45$ K in this study.

3. Capillary pressure is not considered in this work. Thus we only have single fluid pressure field p. Without causing any ambiguity, sometimes p is simply known as the pressure field.

2.2. Porosity Evolution Equations

In this work, we assume the solid-phase decomposition process is related to the fluid pressure p through the following equation:

$$\dot{\phi}^f = k_{sd} \langle p^* - p \rangle (\phi^\infty - \phi^f), \tag{1}$$

where ϕ^f is the porosity and $\dot{\phi}^f$ implies the time derivative, $\phi^\infty \geq \phi^f$ is the limit value of ϕ^f when the decomposition process is completed, k_{sd} is denoted as the solid decomposition constant, p^* is the decomposition equilibrium pressure, and $\langle \circ \rangle$ is the Macaulay bracket notation. If we set $k_{sd} = 0$ or p^* is always be smaller than p, then the porosity ϕ^f would remain a constant and the conventional multiphase fluid flow formulation would apply. Please note that in order to solve Equation (1), the initial value of ϕ^f is also needed.

We could understand Equation (1) by making analogies to the rock pressure solution and hydrate dissociation processes. In rock mechanics, "pressure solution" refers to a deformation mechanism where mineral grains dissolve at their points of contact under high stress conditions [11]. Thus a reduction in the fluid pressure may lead to a heightened effective stress, which could enhance the stress concentration at grain contacts, thereby promoting the pressure solution process. For hydrate, the dissociation happens during depressurization [4]; however, it is important to realize that Equation (1) is essentially different from the so-called Kim-Bishnoi kinetic model [29], as the hydrate dissociation is a strongly endothermic process, which is capable of reducing the temperature by several degrees in a matter of minutes [29,30]. In contrast, the proposed Equation (1) is valid for isothermal formulations. In addition, the Kim-Bishnoi kinetic model is related to the hydrate saturation S_h, while the proposed Equation (1) is related to the porosity ϕ^f (S_h never appears in our formulation).

2.3. Mass Balance Equations

After prescribing the evolution law for porosity ϕ^f, the mass balance equations for the wetting phase and non-wetting phase fluids can be derived as

$$\frac{\partial \left(\phi^f \rho_{nw} S_{nw}\right)}{\partial t} + \nabla \cdot \left(\rho_{nw} \boldsymbol{q}_{nw}\right) = \rho_{ds} \dot{\phi}^f \chi_{nw}, \tag{2}$$

$$\frac{\partial \left(\phi^f \rho_w S_w\right)}{\partial t} + \nabla \cdot \left(\rho_w \boldsymbol{q}_w\right) = \rho_{ds} \dot{\phi}^f \chi_w, \tag{3}$$

where S_w and $S_{nw} = 1 - S_w$ denote the saturations of the wetting and non-wetting phase fluids, respectively. The corresponding Darcy velocities are given by $\boldsymbol{q}_w$ and $\boldsymbol{q}_{nw}$. ρ_{ds} is the decomposable solid density. Finally, χ_w and $\chi_{nw} = 1 - \chi_w$ represent the production fractions of the wetting and non-wetting phases, which are assumed to be constant here.

The Darcy velocities for the non-wetting and wetting phases are defined as

$$\boldsymbol{q}_{nw} = -\frac{k\, k_{rnw}}{\mu_{nw}} (\nabla p - \rho_{nw} \mathbf{g}), \qquad \boldsymbol{q}_w = -\frac{k\, k_{rw}}{\mu_w} (\nabla p - \rho_w \mathbf{g}). \tag{4}$$

In the above equation, the gravity vector $\mathbf{g}$ is set to zero. The absolute permeability k is expressed as

$$k = k_0 \left(1 - \frac{\phi^\infty - \phi^f}{\phi^\infty}\right)^N, \tag{5}$$

where $N = 10$ is the permeability change exponent and k_0 is the reference permeability. Although the viscosity of the non-wetting phase, μ_{nw}, is theoretically a complex function of density and temperature [7], in this work a constant value is adopted for simplicity, and similar treatments are also employed in Klar et al. [31] and Cao et al. [32]. Finally in Equation (4), k_{rw} and k_{rnw} are relative permeabilities, and they are defined as functions of saturations as [7]

$$k_{rnw} = k_{rnw0}\left(\frac{S_{nw} - S_{nwr}}{1 - S_{wr} - S_{nwr}}\right)^{n_{nw}}, \tag{6}$$

$$k_{rw} = k_{rw0}\left(\frac{S_w - S_{wr}}{1 - S_{wr} - S_{nwr}}\right)^{n_w}, \tag{7}$$

where n_w and n_{nw} are relative permeability exponent, $k_{rnw0} = k_{rw0} = 1$ are relative end-point permeabilities, and S_{wr} and S_{nwr} are known as irreducible saturations.

3. Hybrid Explicit-Implicit FEM

In the proposed hybrid explicit-implicit FEM, the calculation is divided into three stages: (I) implicit update of ϕ^f; (II) implicit update of pressure p; (III) explicit update of saturation S_w. Suppose we have $\phi^{f,n}$, p^n, and S_w^n at time step $t = t_n$, through the above three-stage calculation, we will obtain $\phi^{f,n+1}$, p^{n+1}, and S_w^{n+1}. The details are given as follows. Note we always use the superscript to denote the discretized time level of variables. Although intermediate operations are performed at the integration points, all variables ($\phi^{f,n+1}$, p^{n+1}, and S_w^{n+1}) are ultimately stored at the element nodes.

For the first stage, as the pressure field can only take $p = p_n$, Equation (1) could be solved as a first-order ordinary differential equation (ODE), which gives

$$\phi^{f,n+1} = \phi^{\infty} + \left(\phi^{f,n} - \phi^{\infty}\right) \exp\left[-k_{sd}\left\langle p^* - p^n\right\rangle \Delta t\right]. \tag{8}$$

For stage II, we will solve for the "pressure equation" by adding Equation (2) with Equation (3), similarly to that in the implicit pressure explicit saturation method (IMPES) in reservoir simulation. While IMPES is based on the FVM, our method is based on the FEM. The total flow equation could be written as

$$\frac{\partial\left(\phi^f \rho_{nw} S_{nw}\right)}{\partial t} + \frac{\partial\left(\phi^f \rho_w S_w\right)}{\partial t} + \nabla \cdot \left(\rho_{nw}\, \boldsymbol{q}_{nw} + \rho_w \boldsymbol{q}_w\right) = \rho_{ds}\, \dot{\phi}^f, \tag{9}$$

Then we could proceed to derive the finite element equation for pressure in the standard way, and the residual array equation takes the following form (assuming the Dirichlet boundary condition is prescribed at the outlet, and the zero-flux boundary condition is valid for other boundaries):

$$\begin{aligned}\mathcal{R}_p^{n+1} = &\int_\Omega \mathbb{N}_p^{\mathrm{T}} \frac{\phi^{f,n+1}\rho_{nw}^{n+1} S_{nw}^n + \phi^{f,n+1}\rho_w S_w^n - \phi^{f,n}\rho_{nw}^n S_{nw}^n - \phi^{f,n}\rho_w S_w^n}{\Delta t}\, \mathrm{d}V \\ &- \int_\Omega \mathbb{E}_p^{\mathrm{T}} \left(\rho_{nw}^{n+1} \boldsymbol{q}_{nw}^{n+\frac{1}{2}} + \rho_w\, \boldsymbol{q}_w^{n+\frac{1}{2}}\right) \mathrm{d}V \\ &+ \int_\Omega \mathbb{N}_p^{\mathrm{T}} \rho_{ds}\, k_{sd} \left\langle p^* - p^{n+1}\right\rangle \left(\phi^{\infty} - \phi^{f,n+1}\right) \mathrm{d}V,\end{aligned} \tag{10}$$

where $\mathbb{N}_p$ and $\mathbb{E}_p$ are the standard shape function and shape function gradient matrices [33], and the superscript $n + \frac{1}{2}$ implies that the time discretization level is mixed for $\boldsymbol{q}_{nw}$ and $\boldsymbol{q}_w$, given as

$$\boldsymbol{q}_{nw}^{n+\frac{1}{2}} = -\frac{k^{n+1}\, k_{rnw}^n}{\mu_{nw}}\, \mathbb{E}_p\, \mathbf{p}^{n+1}, \qquad \boldsymbol{q}_w^{n+\frac{1}{2}} = -\frac{k^{n+1}\, k_{rw}^n}{\mu_w}\, \mathbb{E}_p\, \mathbf{p}^{n+1}. \tag{11}$$

From the equation above, it is clear that because the saturation field is not updated during the second stage. Consequently, relative permeabilities must be calculated using saturation values from the previous time level, resulting in a mixed time discretization scheme for the Darcy velocities $\boldsymbol{q}_w$ and $\boldsymbol{q}_{nw}$.

Finally, for the stage III, we will update the saturation field S_w (note the S_{nw} field will be updated automatically) through the transport Equation (3) explicitly. As we will reveal through numerical calculations, the update of saturation field must be achieved through the transport equation of an incompressible or nearly incompressible fluid phase, rather than a highly compressible fluid phase such as the gas phase. The discretized counterpart of Equation (3) is given as

$$\frac{\phi^{f,n+1}S_w^{n+1} - \phi^{f,n}S_w^n}{\Delta t} + \nabla \cdot \boldsymbol{q}_w^{n+\frac{1}{2}} = \frac{\rho_{ds}}{\rho_w}\chi_w\, k_{sd}\left\langle p^* - p^{n+1}\right\rangle\left(\phi^{\infty} - \phi^{f,n+1}\right). \tag{12}$$

Here, we again adopt the mixed time discretization for $\boldsymbol{q}_w$. In Equation (12), S_w^{n+1} appears only in the first term on the left-hand side; consequently, this stage is referred to as the explicit update of saturation. Another critical issue is the evaluation of $\nabla \cdot \boldsymbol{q}_w^{n+\frac{1}{2}}$ at three Gauss integration points as well as at element nodes. This is because we adopt the linear triangle (T3) finite element, we cannot directly take the gradient twice, as it would give a trivial zero value. Therefore, we first calculate $\boldsymbol{q}_w^{n+\frac{1}{2}}$ at three Gauss integration points using Equation (11). In the next step, we map $\boldsymbol{q}_w^{n+\frac{1}{2}}$ from three integration points to nodes according to the procedure given in Appendix A. Let us denote the nodal Darcy velocity as $\widetilde{\boldsymbol{q}}_w^{n+\frac{1}{2}}$, and if we concatenate all the nodal Darcy velocities into a $n_{\text{node}} \times 2$ matrix where n_{node} is the number of nodes in a FEM mesh, we could represent $\nabla \cdot \boldsymbol{q}_w^{n+\frac{1}{2}}$ at Gauss integration points as

$$\nabla \cdot \boldsymbol{q}_w^{n+\frac{1}{2}} = \mathrm{Tr}\left[\mathbb{E}_p\, \widetilde{\boldsymbol{q}}_w^{n+\frac{1}{2}}\right], \tag{13}$$

where $\mathrm{Tr}[\cdot]$ represents the trace operator of a square matrix. Lastly, we need to re-map $\nabla \cdot \boldsymbol{q}_w^{n+\frac{1}{2}}$ from integration points to nodes using exactly the same procedure given in Appendix A, then we could apply Equation (12) to update the saturation field at element nodes. This re-mapping procedure is sometimes necessary, especially in the latter Buckley-Leverett problem, as we need to ensure S_w at inlet boundary nodes remains at 1.

However, simulations suggest that a pure update using Equation (12) cannot provide satisfactory results, and as a result, Equation (12) should be modified to add an artificial diffusion term [34], and Equation (12) becomes

$$\frac{\phi^{f,n+1}S_w^{n+1} - \phi^{f,n}S_w^n}{\Delta t} + \nabla \cdot \boldsymbol{q}_w^{n+\frac{1}{2}} = \frac{\rho_{ds}}{\rho_w}\chi_w\, k_{sd}\left\langle p^* - p^{n+1}\right\rangle\left(\phi^{\infty} - \phi^{f,n+1}\right) + \underbrace{\delta h|\boldsymbol{q}_T|}_{c_{art}}\nabla^2 S_w^n, \tag{14}$$

where c_{art} is the artificial diffusion coefficient, in the dimension of $[\mathrm{L}]^2/[\mathrm{T}]$, δ is a dimensionless tunable parameter whose suggested value is 0.25 by COMSOL Multiphysics (version 6.3) [35]; h is the characteristic mesh size; and $|\boldsymbol{q}_T|$ is the total Darcy velocity magnitude, which could be estimated using the inlet and outlet pressures, characteristic fluid mobility, and model dimension in the flow direction. The evaluation of $\nabla^2 S_w^n$ follows a similar recipe as before because $\nabla^2 S_w^n = \nabla \cdot (\nabla S_w^n)$.

4. Model Simulation and Verification

4.1. Multiphase Flow Under Depressurization

We consider a two-dimensional rectangular domain, as illustrated in Figure 1.

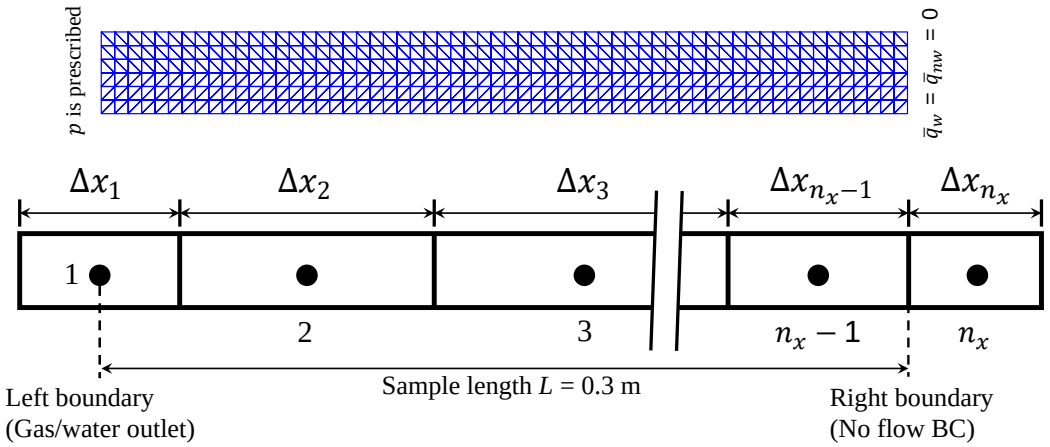


Figure 1. Two-dimensional triangular finite element mesh (in blue color) and the one-dimensional finite volume discretization (block-centered grid) of the proposed depressurization problem. In the FVM, grid 2 to grid $n_x - 1$ would enter the FVM discretization. This is because for grid centroid 1, the pressure is prescribed, and the saturation at this centroid will not enter the FVM residual, so we assume $(S_w)_1 = (S_w)_2$. For grid centroid n_x, since we have a zero-flux boundary condition, we simply assume $(p)_{n_x} = (p)_{n_x-1}$ and $(S_w)_{n_x} = (S_w)_{n_x-1}$.

The system is initialized with uniform pressure, porosity, and saturation fields. At $t = 0^+$, the pressure at the left boundary is instantaneously reduced, a condition that may trigger the mechanism described in Equation (1). Consequently, transient flow is initiated for both the wetting and non-wetting phase fluids, leading to an evolution of the saturation fields. While many prior studies provide only vague descriptions of boundary conditions, precise definitions are essential for algorithmic implementation; therefore, we detail the specific boundary conditions used in this study in Figure 1. As no analytical solution exists for this problem, verification is performed by comparing our results with those obtained using a fully implicit FVM [1]. We examine cases involving both constant porosity and variable porosity resulting from solid-phase decomposition and phase change.

4.1.1. Constant Porosity Case (No Solid Decomposition)

We first analyze the case with a constant porosity $\phi^f = 0.182$ and all the other material parameters are provided in Table 1.

Due to the depressurization at the left end, the pressure gradient drives the existing fluids within the porous medium toward the left boundary, where they exit the domain. Meanwhile, the right boundary is defined as a zero-flux Neumann boundary, which prevents any fluid replenishment. Consequently, the mass terms $\phi^f \rho_w S_w$ and $\phi^f \rho_{nw} S_{nw}$ are both expected to decrease. For the wetting phase fluid, since ρ_w is a constant, so the outflow of the wetting phase necessitates a decrease in saturation S_w, which in turn causes an increase in S_{nw} (since $S_w + S_{nw} \equiv 1$). This raises a question: given that the non-wetting phase fluid also flows out of the domain, why does its saturation increase rather than decrease? The explanation lies in the compressibility of the non-wetting phase fluid; its density decreases as pressure drops. Therefore, the overall product $\phi^f \rho_{nw} S_{nw}$ still decreases, which remains consistent with physical expectations.

Figures 2 and 3 present the simulation results for both the proposed hybrid explicit-implicit FEM and the FVM. The saturation evolution in Figure 2 is consistent with the aforementioned analysis, and the results from both methods demonstrate good agreement. Regarding the pressure field, the system compressibility arising from the non-wetting phase ensures that the pressure equation remains parabolic. Consequently, it takes time for the pressure field to decay until it reaches a uniform state equal to the outlet pressure,

which is shown in Figure 3. In other words, suppose our non-wetting phase fluid is also incompressible, we would expect the pressure decay to finish immediately, while the saturation throughout the entire domain would remain unchanged; the fluid production from the outlet is also zero. In fact, this is indeed tested in Figure 4, in which we assume $\mu_{nw} = \mu_w = 0.001$ Pa $\cdot$ s and the non-wetting phase fluid density is described using the following function

Table 1. Material parameters of the porous medium utilized in the multiphase flow numerical model accounting for depressurization.

Parameter	Value	Unit
Porosity evolution:		
Constant porosity $\phi^f = \phi^\infty$	0.182	1
Hydraulic:		
Constant permeability k_0	10^{-15}	m^2
Non-wetting phase fluid viscosity μ_{nw}	2×10^{-5}	Pa·s
Wetting phase fluid viscosity μ_w	0.001	Pa·s
Relative permeability exponents n_w and n_{nw}	1.5, 2	1
Irreducible saturations S_{wr} and S_{nwr}	0, 0	1
Initial wetting phase saturation S_w	0.3	1
Others:		
Constant temperature Θ	275.45	K
Model length	0.3	m
Model width	0.03	m
Initial fluid pressure	3.75	MPa
Outlet fluid pressure	1.25	MPa
Time step size Δt	1	s
Number of time steps	1000	1
Characterisic mobility in estimating $\lvert \boldsymbol{q}_T \rvert$ of (14)	10^{-11}	$m^2/Pa/s$
Coefficient δ in Equation (14)	0.25	1

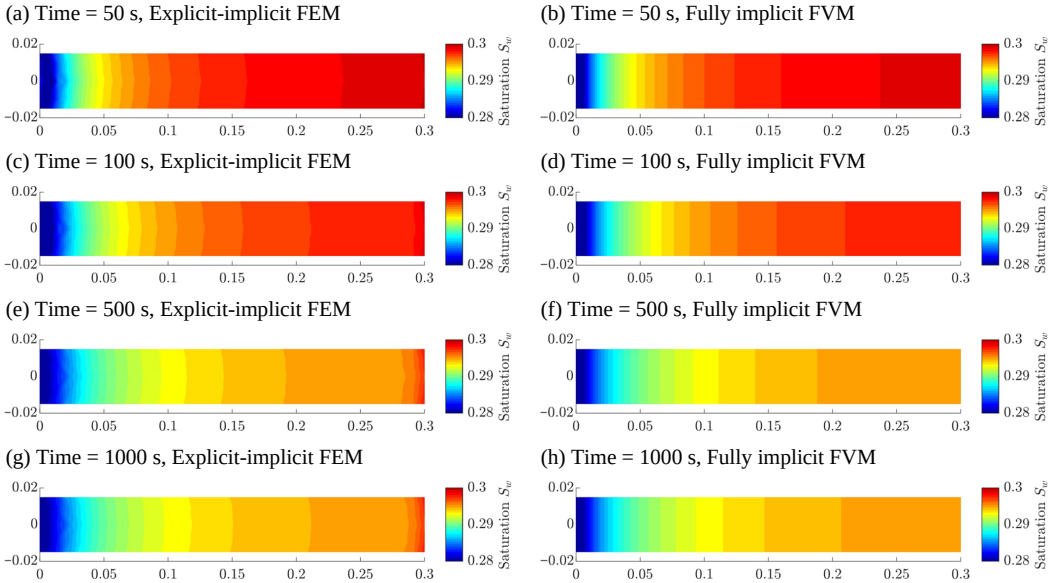


Figure 2. Comparison of saturation distributions at different time instants between the explicit-implicit FEM and fully implicit FVM. The results from both methods are generally in good agreement. However, at later stages, the FEM solution exhibits a slight accumulation of high S_w saturation at the right boundary. This discrepancy may be attributed to the lack of local mass conservation in the proposed method.

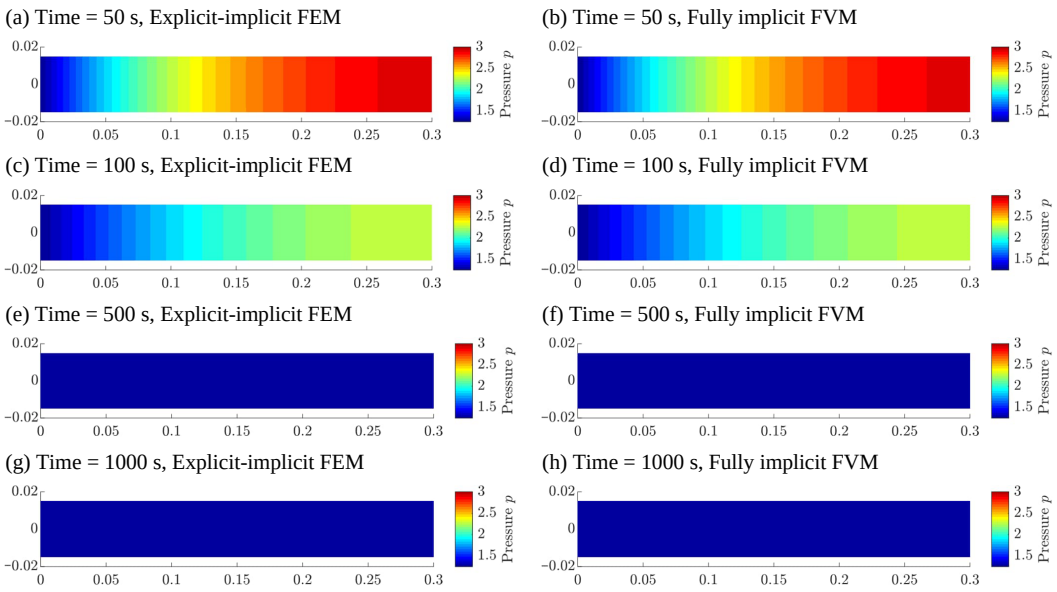


Figure 3. Comparison of pressure distributions at different time instants between the explicit-implicit FEM and fully implicit FVM. It takes time for the pressure field to decay until it reaches a uniform state equal to the outlet pressure.

$$\rho_{nw} = \rho_{nw}^{\text{ref}} \left(1 + \frac{p}{K_{nw}}\right). \tag{15}$$

In the above equation, $\rho_{nw}^{\text{ref}} = 900\ \text{kg/m}^3$ is the reference density, and K_{nw} is the non-wetting phase fluid bulk modulus, which is assumed to be a very large number (we use 2×10^{15} Pa here). Now we can see that even for such a small time step ($\Delta t = 0.001$ s), the pressure equilibrium is reached instantaneously, with negligible saturation changes.

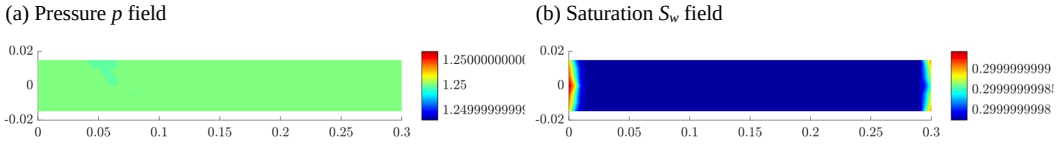


Figure 4. Due to the system incompressibility of both the wetting phase and non-wetting phase fluids, the pressure equilibrium is reached instantaneously for an arbitrarily small time step ($t_{\text{total}} = 0.01$ s for 10 uniform time steps), and there is no saturation change.

In Section 3, we have mentioned that the update of saturation field must be achieved through the transport equation of the incompressible or nearly incompressible fluid phase, rather than the highly compressible fluid phase such as the gas phase. This statement is actually derived here. In Figure 5, we show the S_w distribution just after one time step if we use Equation (2) in stage III, where obvious non-realistic negative saturation values could be observed.

4.1.2. Variable Porosity Case (With Solid Decomposition and Phase Change)

We next investigate the case with solid-phase decomposition and phase change, and the material parameters are provided in Table 2.

As one can see from the table, due to the consideration of solid-phase decomposition and the production of fluid phases, a smaller time step size and a larger number of total time steps are needed, compared with the previous example with constant porosity. Figures 6 and 7 plot the results of ϕ^f and S_w.

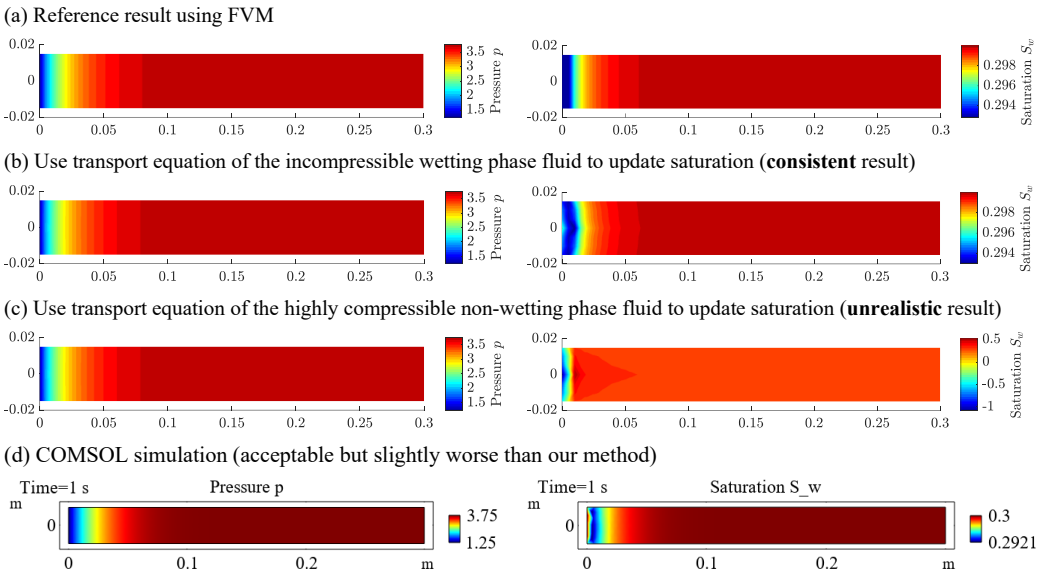


Figure 5. Comparison of different saturation updating strategies in stage III. The simulation just runs for one time step with $\Delta t = 1$ s.

Table 2. Material parameters of the porous medium utilized in the multiphase flow numerical model accounting for depressurization and variable porosity (due to solid-phase decomposition).

Parameter	**Value**	**Unit**
Porosity evolution:		
Initial porosity ϕ^f	0.0908	1
Limit value ϕ^∞	0.182	1
Solid decomposition constant k_{sd}	$1.54649789 \times 10^{-9}$	1/Pa/s
Decomposition equilibrium pressure p^*	3.678778	MPa
Decomposable solid density ρ_{ds}	910	kg/m^3
Production fraction χ_w ($\chi_{nw} = 1 - \chi_w$)	0.12923	1
Hydraulic:		
Reference permeability k_0	$9.6698745 \times 10^{-14}$	m^2
Permeability change exponent N	10	1
Non-wetting phase fluid viscosity μ_{nw}	2×10^{-5}	Pa·s
Wetting phase fluid viscosity μ_w	0.001	Pa·s
Relative permeability exponents n_w and n_{nw}	4, 2	1
Irreducible saturations S_{wr} and S_{nwr}	0.1, 0	1
Initial wetting phase saturation S_w	0.7034	1
Others:		
Constant temperature Θ	275.45	K
Model length	0.3	m
Model width	0.03	m
Initial fluid pressure	3.75	MPa
Outlet fluid pressure	2.84	MPa
Time step size Δt	0.1	s
Number of time steps	60,000	1

The porosity evolution trend given by Figure 6 is quite consistent with our intuition. Since depressurization is applied to the left boundary, decomposition is initiated locally, accompanied by a rapid rise in porosity. Subsequently, porosity levels across the entire region are observed to increase, eventually stabilizing at the limiting value ϕ^∞. Regarding the saturation field, a slight decrease is observed at the left boundary, which gradually

propagates to the right. However, the overall variation in S_w is minimal (approximately 4.3%), especially when compared to ϕ^f, which doubles in value.

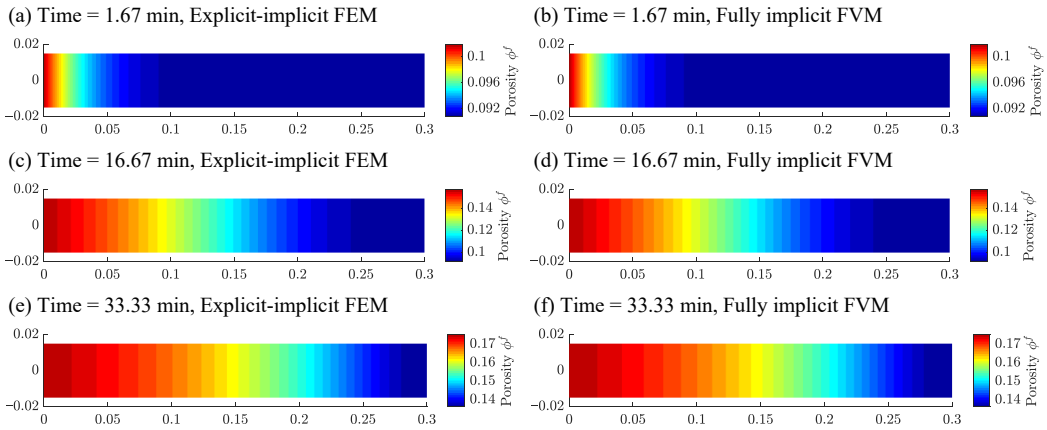


Figure 6. Comparison of variable porosity distributions at different time instants between the explicit-implicit FEM and fully implicit FVM considering solid-phase decomposition and phase change.

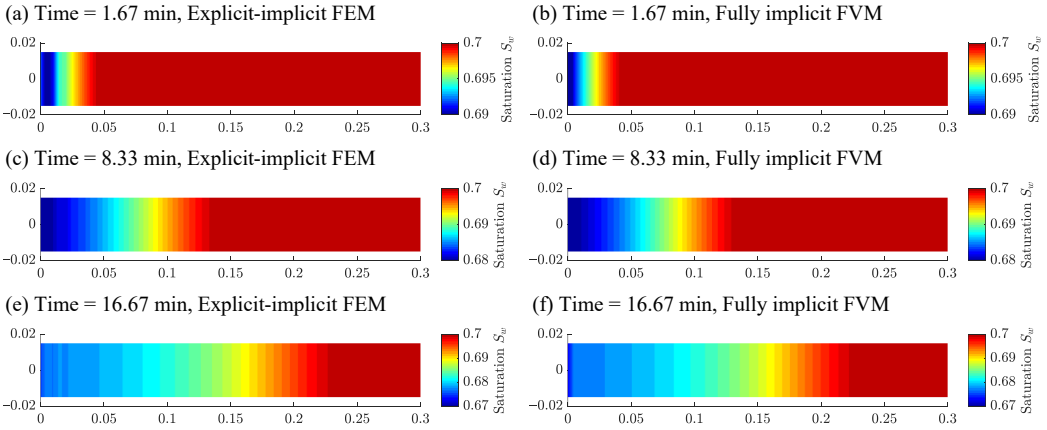


Figure 7. Comparison of saturation distributions at different time instants between the explicit-implicit FEM and fully implicit FVM considering solid-phase decomposition and phase change.

For the evolution of pressure at the right boundary, as shown in Figure 8a, we could visually divide the pressure decay into five stages. In order to capture the first stage, which is also the stage with the shortest duration, the time step size cannot be very large. In addition, in the third stage, the curve is almost flat. This kind of behavior has only been reported from a double porosity system [33,36–39] due to the different drainage capacities of macropores and micropores, but as we have shown here, it could also appear in the reactive porous media.

Finally, the comparison of cumulative gas production is also critical to ensure the effectiveness of our proposed approach, and in fact, the cumulative gas production curve almost appears on every numerical study of gas reservoirs and MHBS. Differently to the FVM which naturally ensures the local mass conservation, so the gas production could be accumulated from the flux across the face between the first and the second cell (see Figure 1), for the FEM-based method, we will calculate the cumulative gas production using the material balance law. Specifically, we will count the gas production through the source term in Equation (2), as well as the remaining free gas mass in the pores. We also need to account for the change of geometry, as in Figure 8b, the geometry is a cylinder with $L \times R = 0.3\,\text{m} \times 0.0255\,\text{m}$ (L is the cylinder length and R is the cylinder cross-section radius), though the flow is still one-dimensional. Figure 8b shows the cumulative gas production results obtained from two numerical methods. An excellent match can be

observed. Another interesting observation regarding the gas cumulative production curve is that, in its early stage, the curve is convex, only transitioning to a concave shape after 20 min. This behavior differs from that of conventional shale gas production curves that are always concave when plotting using the linear time axis [1,40], and we preliminarily attribute the difference to the distinct gas production mechanisms.

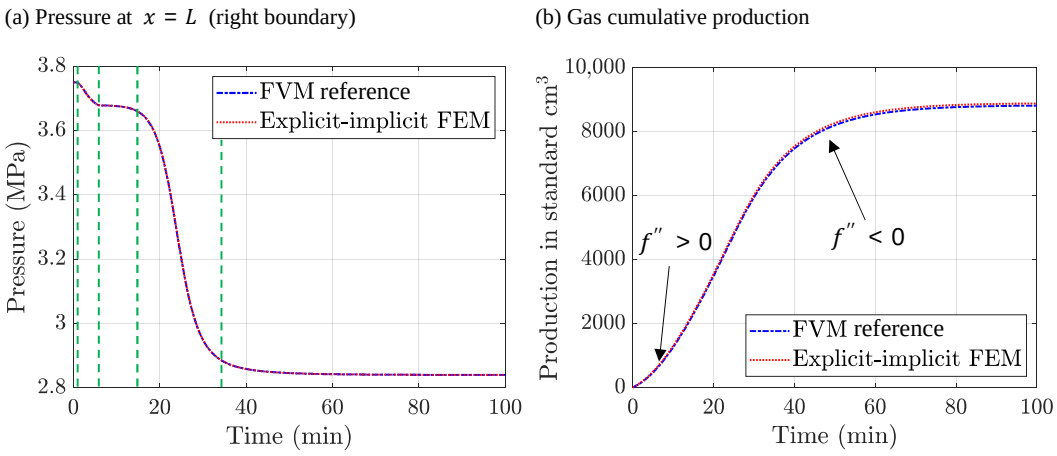


Figure 8. Comparison of (**a**) pressure at the right boundary ($x = L$, $y =$ width/2) and (**b**) gas cumulative production between the explicit-implicit FEM and fully implicit FVM.

4.2. Buckley-Leverett Problem

We finally apply the same numerical method to a two-phase immiscible fluid displacement problem, which is also known as the Buckley-Leverett problem [41]. Through this example, we aim to show the necessity of the correction term in Equation (14).

The model is sketched in Figure 9 and the material parameters are provided in Table 3. Both the wetting phase and non-wetting phase fluids are assumed to be incompressible. The analytical solution of the Buckley-Leverett problem could be sought from Rodriguez-Torrado et al. [28] and Fuks and Tchelepi [41]. For the parameters mentioned in Table 3, the shock location is at the saturation $S_w^* = \sqrt{2}/2$ and the dimensionless shock wave velocity is $v_{shD} = (\sqrt{2}+1)/2$.

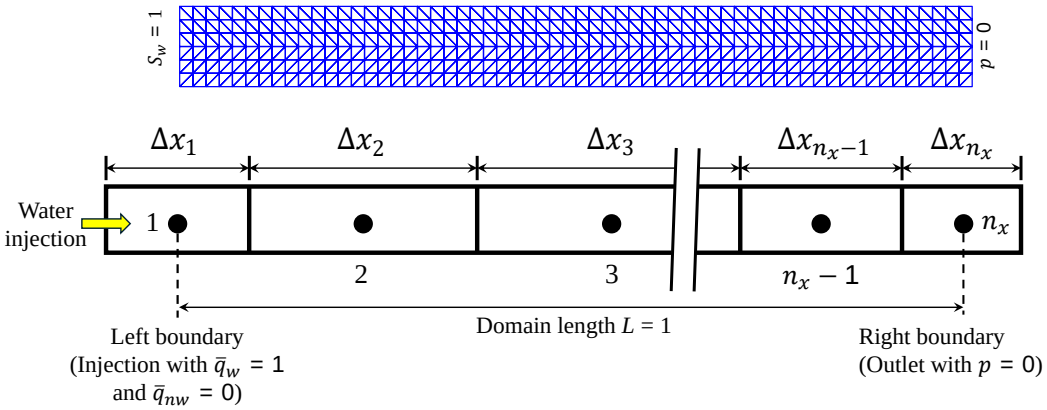


Figure 9. Two-dimensional triangular finite element mesh (in blue color) and the one-dimensional finite volume discretization (block-centered grid) of the Buckley-Leverett problem. In the FVM, grid 1 to grid $n_x - 1$ would enter the FVM discretization. This is because for grid centroid n_x, the pressure is prescribed at 0, and we assume $(S_w)_{n_x} = (S_w)_{n_x-1}$. For grid centroid 1, the Darcy flux of the wetting phase fluid for the left face is 1, and 0 for the non-wetting phase fluid.

Table 3. Material parameters of the porous medium utilized in the Buckley-Leverett problem. Please note that all the variables in the Buckley-Leverett problem are dimensionless, so we drop the "unit" column.

Parameter	**Value**
Porosity evolution:	
Constant porosity $\phi^f = \phi^\infty$	1‡
Hydraulic:	
Constant permeability k_0	1
Non-wetting phase fluid viscosity μ_{nw}	1
Wetting phase fluid viscosity μ_w	1
Relative permeability exponents n_w and n_{nw}	2, 2
Irreducible saturations S_{wr} and S_{nwr}	0, 0
Initial wetting phase saturation S_w	0
Others:	
Model length	1
Model width	0.1
Initial fluid pressure	0
Outlet fluid pressure on the right boundary	0
Injection total flux	1
Injection wetting phase fraction	1
Time step size Δt	0.001
Number of time steps	1000

‡ This is not a typo. Since we are solving a dimensionless PDE, and the porosity will be absorbed into the dimensionless shock velocity, so setting porosity to 1 is totally acceptable.

Figure 10 depicts the results of water saturation and pressure using Equation (12). Unfortunately, the numerical solution fails to adequately capture the rarefaction wave, and the predicted shock front lags behind its true position. Regarding the pressure field, pronounced differences are observed only in the region between the inlet and the shock front, i.e., the region that has already been flooded by the wetting phase fluid.

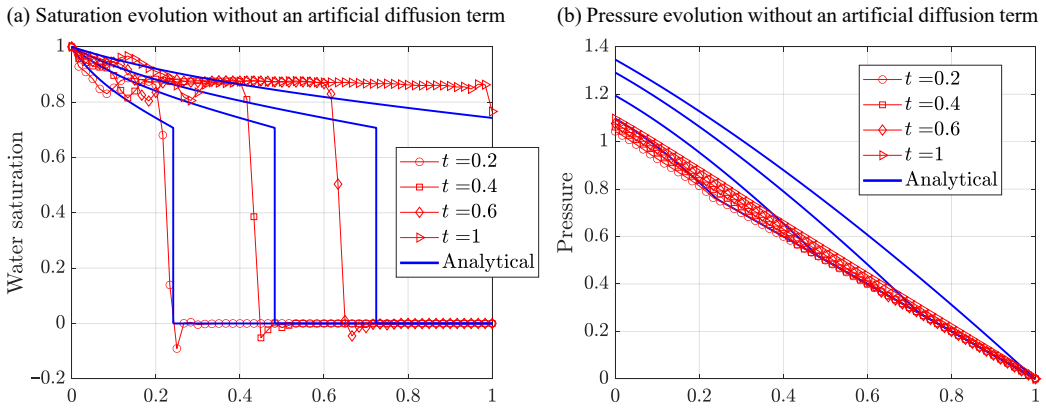


Figure 10. Comparison between the numerical results and analytical solutions for the Buckley-Leverett problem in the absence of an artificial diffusion term. Note the significant deviations in the upstream part (rarefaction wave), and the predicted shock front also lags behind its true position.

In contrast, Figure 11 shows the updated results after incorporating the artificial diffusion term with $\delta = 0.375$, $h = 1/60$, and $|q_T| = 1$. Now we could observe that both the rarefaction wave and the shock wave could be correctly captured even after the breakthrough $t > 1/v_{shD}$. By comparing with the FVM results in Figure 12, our proposed method, even when not outperforming FVM, performs at least as well as FVM.

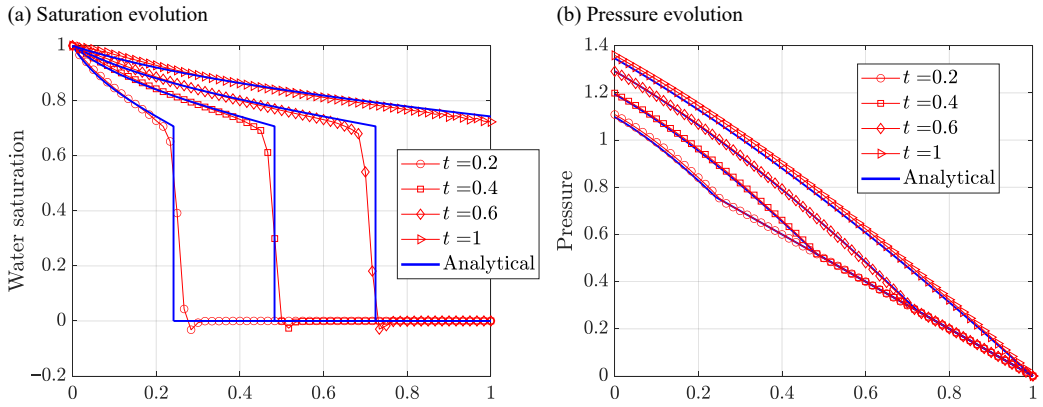


Figure 11. Consistent comparison with the analytical solution at various time instants after incorporating an artificial diffusion term, see Equation (14). Our proposed method could naturally deal with the stage after breakthrough, while for fully implicit FEM, the non-zero wetting phase fluid flux at the right boundary after breakthrough must be considered, in order to obtain reasonable results.

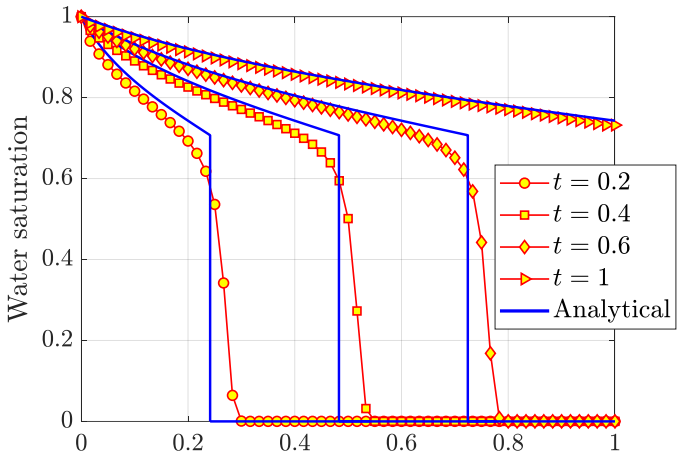


Figure 12. Saturation evolution from the fully implicit finite volume method. In order to have a fair comparison with Figure 11, in the FVM, we take $\Delta x = 1/60$ (same as the triangle element edge length in Figure 9) and $\Delta t = 0.001$ (same time step as FEM). Based on the comparison results, our proposed method, even when not outperforming FVM, performs at least as well as FVM.

5. Conclusions

Porous media multiphase flow is ubiquitous in many physical processes, and some of these processes also involve solid-phase decomposition, such as hydrate dissociation and internal erosion, yet the complexity remains challenging to model consistently. As revealed by recent code-comparison studies, there are still evident inconsistencies in the numerical results for the same benchmark problem across different research groups. In response, this study introduces a new, simple, and reliable benchmark test and presents a hybrid explicit-implicit finite element method that is adaptable to a wide range of scenarios. In our mathematical framework, the solid decomposition is described by a rate equation for porosity ϕ^f that depends on the fluid pressure p, and the phase change is modeled via mass source terms. It is definitely possible to replace the current rate equation of ϕ^f by other empirical laws, and, as long as we can perform the numerical integration to get $\phi^{f,n+1}$ from $\phi^{f,n}$, our method still applies. Detailed descriptions of the methodology are provided, including the innovative three-stage updating strategy, incorporation of an artificial diffusion term, and the selection of the transport equation in the final stage. The numerical results from our method are rigorously validated against those from the fully implicit finite volume method and analytical solutions.

Validation results indicate that, despite minor discrepancies at the boundaries, the finite volume method and our proposed method are in substantial agreement, thereby demonstrating the reliability of our approach. Notably, in the Buckley-Leverett problem, our method yields results closer to the analytical solution than the FVM under identical mesh sizes and time steps, although slight overshoots and undershoots in saturation were observed. Furthermore, the analysis confirms that the saturation update must utilize the transport equation for the incompressible (or nearly incompressible) fluid phase. It is also evident that the artificial diffusion term is critical to capturing a physically correct saturation profile; therefore, this correction should generally be retained, even when advection is not dominant. Future work may explore locally mass-conservative finite element frameworks [42–44] to investigate whether they can further mitigate the observed discrepancies in the saturation profile. The impact of anisotropy will also be investigated [45–48].

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Data Availability Statement: The raw data supporting the conclusions of this article will be made available by the authors on request.

Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A. Recovery of Nodal Variables from Values at Integration Points

For each linear triangle element (three nodes), we first map the values at three integration points to element nodes (see Figure A1 below) using the following linear equation:

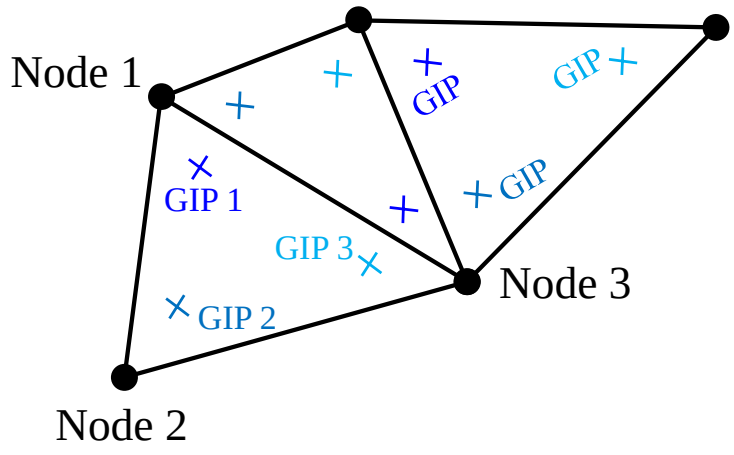


Figure A1. Schematic diagram of recovery of nodal variables from values at **G**auss **I**ntegration **P**oints (GIP 1, GIP 2, GIP 3).

$$\begin{pmatrix} \xi^1_{\text{Node}} \\ \xi^2_{\text{Node}} \\ \xi^3_{\text{Node}} \end{pmatrix} = \begin{pmatrix} 2/3 & 1/6 & 1/6 \\ 1/6 & 2/3 & 1/6 \\ 1/6 & 1/6 & 2/3 \end{pmatrix}^{-1} \begin{pmatrix} \xi^1_{\text{GIP}} \\ \xi^2_{\text{GIP}} \\ \xi^3_{\text{GIP}} \end{pmatrix}, \tag{A1}$$

where ξ denotes any scalar field. In practice, if the physical quantity under consideration is a vector (e.g., the Darcy velocity $\mathbf{q}_w$) or a second-order tensor (such as the stress tensor σ), the above transformation is applied to each of its components individually. After performing this mapping, each node accumulates several values from the adjacent elements (i.e., every triangle sharing that node). For instance, as shown in Figure A1, Node 3 might be connected to three triangles ($n_{\text{connele}} = 3$) while Node 1 may be associated with only two ($n_{\text{connele}} = 2$). A final nodal value for each variable is then obtained by computing a weighted average of these contributions, with the weights determined by the areas of the connected triangles. This completes the recovery process.

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